



Regular article

Diagnosis of the three-phase induction motor using thermal imaging

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HIGHLIGHTS

- Techniques of fault diagnosis of three-phase induction motors are proposed.
- Original method of feature extraction of thermal images called MoASoID is described.
- Three states of the three-phase induction motor are analysed.
- Efficiency of recognition of thermal image is analysed.

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ABSTRACT

Three-phase induction motors are used in the industry commonly for example woodworking machines, blowers, pumps, conveyors, elevators, compressors, mining industry, automotive industry, chemical industry and railway applications. Diagnosis of faults is essential for proper maintenance. Faults may damage a motor and damaged motors generate economic losses caused by breakdowns in production lines. In this paper the authors develop fault diagnostic techniques of the three-phase induction motor. The described techniques are based on the analysis of thermal images of three-phase induction motor. The authors analyse thermal images of 3 states of the three-phase induction motor: healthy three-phase induction motor, three-phase induction motor with 2 broken bars, three-phase induction motor with faulty ring of squirrel-cage. In this paper the authors develop an original method of the feature extraction of thermal images MoASoID (Method of Areas Selection of Image Differences). This method compares many training sets together and it selects the areas with the biggest changes for the recognition process. Feature vectors are obtained with the use of mentioned MoASoID and image histogram. Next 3 methods of classification are used: NN (the Nearest Neighbour classifier), K-means, BNN (the back-propagation neural network). The described fault diagnostic techniques are useful for protection of three-phase induction motor and other types of rotating electrical motors such as: DC motors, generators, synchronous motors.

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1. Introduction

A number of operating electric motors increases every month [1]. Three-phase induction motors are used in the industry commonly for example woodworking machines, blowers, pumps, conveyors, elevators, compressors, mining industry, automotive industry, chemical industry, railway applications. Often electric motors are required to run continuously without interruption. It is essential in the industry. An early diagnosis of faults prevents

financial losses and unplanned breakdowns in production lines. Faults of electrical rotating motors can be listed as: stator faults, rotor electrical faults, rotor mechanical faults [1]. However, techniques of mentioned faults are becoming complex and specific [2].

In the literature, fault diagnostic techniques were developed by scientists [1–27]. The analyses of electric signals, acoustic signals, vibrations, thermal imaging of electrical motors were very popular in the recent literature. Fault diagnostic techniques of electric currents of electrical motors were developed in the literature [3–6]. MCSA (Motor Current Signature Analysis) is simple technique of fault diagnosis. In the literature, MCSA was used for detecting specific electrical and mechanical induction motor faults [2,3,7]. MCSA has many advantages because it is non-intrusive. It needs

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the stator winding, not affected by the load or asymmetries. There is also simplicity of installation of current sensors. However it has disadvantages, such as the stator current data should be sampled after motor speed arrives at the steady state [3].

Vibration signals were also very popular for fault diagnosis [8–11]. Techniques based on vibration analysis have the advantage of being more sensitive than the MCSA technique for certain faults [2]. These techniques were used to detect specific faults in motors, for example diagnosis of bearings [9] or gear transmission systems [10]. Techniques based on vibration signals give a fault diagnosis focused on the location of spectral components associated with faults [8]. An article presents a comparative study of different time frequency analysis methodologies that can be used for detecting faults in induction motors analysing vibration signals during the startup transient. The analysed vibration signals were one broken rotor bar, two broken bars, unbalance, and bearing defects. The methodology proposed in mentioned paper is applicable when it does not have current signals and only has vibration signals [8]. A methodology based on vibration analysis for the diagnosis of different levels of uniform wear in a gearbox and the detection of bearing defect both linked to the same kinematic chain was presented [9]. The vibration analysis results showed that the amplitude increase in the mesh frequency component and its corresponding harmonics were related to the presence of wear in the gearbox. A review of recent progress of diagnosis of hybrid failures in gear transmission systems using vibration sensor signal was presented [10,11].

The researchers used also acoustic signals for fault diagnosis of machines [12–17]. Techniques based on acoustic signal analysis have the advantage of being cheaper than the other methods. A condenser microphone costs about 30\$. It is also non-invasive. However it has disadvantage. When the induction motor is running a variety of acoustic signals mix together. Reflected and interferenced waves are mixed together. Unfortunately it is very difficult to processing. To solve this problem some feature extraction methods were developed: such as MSAF, MSAF-RATIO, SMOFS and modifications of these methods [12,13]. The results of these methods were similar. Mentioned methods MSAF, SMOFS are good for small training set. For bigger training set it is better to use modifications of these methods: MSAF-RATIO30-MULTIEXPANDED, SMOFS-25-EXPANDED. Another approach for identifying technical condition of combustion engine was described [14,15]. The authors of these papers proposed decision method for identifying engine failure based on sound emission spectral characteristics. It allows fast identification of specific engine damage. Methodology for fault detection in induction motors via sound and vibration signals was showed [16]. The authors of this paper demonstrated the feasibility of detecting faults, such as broken rotor bars, defects in bearings or mechanical unbalance, in induction motors operating at steady-state by analysing acoustic sound signals. The proposed methodology could be used to identify other types of faults in induction motors and in any equipment in which the sound and vibrations were emitted as an unequivocal consequence of malfunctioning. However, the proposed methodology had some limitations. The localization of fault frequencies depended on the knowledge of the motor speed [16]. A bearing fault diagnosis using acoustic emission sensors was presented [17]. The presented technique was validated using the acoustic signals of seeded fault steel bearings on a bearing test rig. The presented approach can be utilized to effectively extract condition indicators to diagnose all four bearing fault types at multiple low shaft speeds below 10 Hz.

Recently fault diagnostic techniques based on thermal imaging gained noticeable attention [18–27]. Thermal imaging is considered for improving the monitoring of induction motors with the advantage of being a non-invasive technique [7]. It can be noticed that thermal images are related with faults for example short-

circuit of stator coils. The analysis of thermal images can be difficult. It is caused by many problems such as: access to many electrical motors, many different parameters of machines (power, size, rotor speed), cooperation with the industry and machine operators, database of thermal images. The researchers should know how to select faults of machine. Moreover long durations of shorted coils can cause permanent damage to the analysed machine. The average time to heat up stator coils is about 1–3 min. Sometimes after such duration stator (rotor) coils can be damaged permanently. For these reasons, the authors decided to analyse fault diagnostic technique of three-phase induction motors (Fig. 1).

In the literature there are not many analyses of early fault diagnostic techniques based on thermal imaging. Diagnostics of electric equipments by means of thermovision was developed [18]. The authors of this paper compared the results of calculated and measured values of breakers in electric switchgear. An analysis of thermal imaging for bearing fault detection of induction motors was also conducted [19]. It was conducted that a faulty bearing of induction motor can be detected using a low-cost camera. It was necessary to consider ambient temperature because absolute thermogram was not enough to determine a faulty state. A detection of induction motor inter turn fault using infrared thermographic analysis was presented [20]. The proposed method used transient thermal monitoring during the start of motor. Next it applied pseudo colouring technique on infrared image of the motor, after it reached a thermal steady state. The monitoring system for three phase induction motor using thermal imaging was implemented [21]. The colour based segmentation technique was used to identify abnormal hot regions in the thermal image. A changing red colour intensity method was implemented to recognise the hot spots.

The NSCT-based (nonsubsampling contourlet transform) infrared image enhancement method for rotating machinery fault diagnosis was developed [22]. The method was experimentally validated in the analysis of rotor test stand in the laboratory. The results of image recognition were in the range of 83–92%. 3D temperature field analysis of the induction motors with broken bar fault was discussed [23]. The paper dealt with 3D temperature estimation for totally enclosed fan cooled (TEFC) induction motor. The authors of this paper analysed the influence of the broken bar number on the motor thermal field. Unsupervised sparse pattern diagnostic of defects with thermography imaging system was developed [24]. The method was based on a thermal mechanism with signal processing-based pattern extraction algorithm using sparse greedy-based principal component analysis (SGPCA). Man-made and natural defects from industry were analysed by the proposed method.

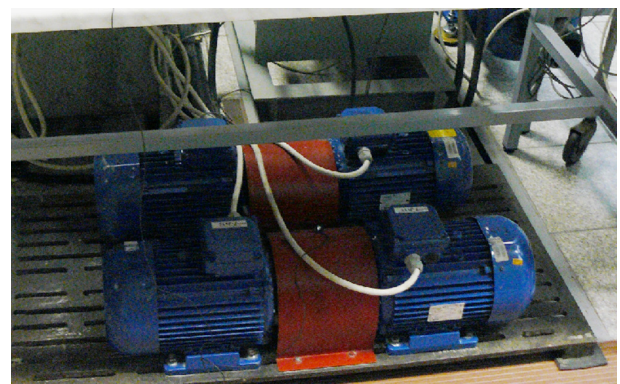


Fig. 1. Four three-phase induction motors (healthy motor, motor with 2 broken coils, motor with faulty ring of squirrel-cage).

The infrared thermal imaging system of ore stockpiles was showed [25]. The authors of the paper conducted on-site and laboratory experiments by using five indices to assess the oxidation characteristics of sulphide ores. A fire source detection method, which can monitor the highest temperature point on the surface of self-igniting ores was also proposed. Thermal imaging of hydro-electric generator using a DTS (Distributed temperature sensing) system was described [26]. The method was based on distributed temperature optical fiber sensing using Raman scattering. The obtained thermal images showed a great potential for monitoring in the stator according to the generator load. Infrared thermography in energy audit of electrical installations was showed [27]. The article provided an overview of the standards in the field of thermal imaging and safety criteria of electrical installations. Methods of face localization in thermograms were proposed [28]. The paper described an algorithm for determination of the head centre in thermograms. Similar methods can be used for induction motor analysis.

In this paper the authors develop an original method of the feature extraction of thermal images MoASoID (Method of Areas Selection of Image Differences). Feature vectors are obtained with the use of mentioned MoASoID and image histogram. Next 3 methods of classification are used: NN (the Nearest Neighbour classifier), K-means, BNN (the back-propagation neural network). The presented fault diagnostic techniques are related to rotor electrical faults. The authors analyse thermal images of 3 states of the three-phase induction motor: healthy three-phase induction motor, three-phase induction motor with 2 broken bars, three-phase induction motor with faulty ring of squirrel-cage. The advantage of proposed technique is related to the method of feature extraction MoASoID. This method compares many training sets together and it selects the areas with the biggest changes for recognition process. In the presented analysis 4 training sets are used. Each training set has 3 thermal images. In the literature the authors proposed MoASoS (Method of Area Selection of States), but this method used only 1 training set [43]. The proposed method of feature extraction MoASoID is extension of MoASoS. It is also used for other type of electrical motor - three-phase induction motor.

2. Proposed fault diagnostic techniques

The proposed fault diagnostic techniques are presented as a block diagram in Fig. 2. Techniques have following steps: acquisi-

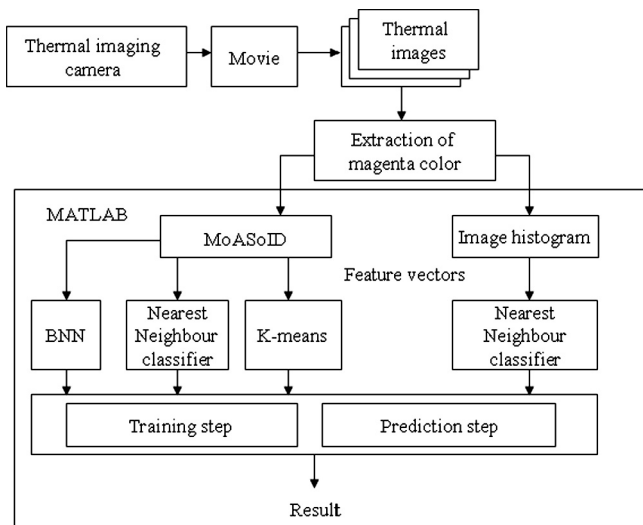


Fig. 2. Block diagram of proposed fault diagnostic techniques.

tion of thermal image, preprocessing, feature extraction and classification. At the beginning of recognition process thermal images are taken with the application of thermal imaging camera FLIR E4 (Temperature Range: -20 to 250 °C, Frame Rate: 9 Hz, Detector: Uncooled Microbolometer) and PC software FLIR IR Camera Player. On the market there are many available thermal imaging cameras with similar or better parameters. These cameras can be also used for the analysis.

Mentioned thermal imaging camera is placed 0.25 m from the analysed motor (Fig. 3).

Movies and thermal images are acquired with application of PC software and thermal imaging camera. Thermal images have resolution of 320×240 pixels (Figs. 4–6). Next extraction of magenta colour is calculated. After that feature extraction is performed with the application of the MoASoID and the image histogram. Next classification of obtained feature vectors is performed by using NN (the Nearest Neighbour classifier), K-means, BNN (the back-propagation neural network).

2.1. Colour extraction

In thermal images, colour extraction is essential task, because some colours are more important than others. The most important colours were white, magenta and red for analysed images. The highest temperatures had white colour. The authors decided to analyse magenta and white. Magenta and white colours of thermal images of the three-phase induction motor are obtained by using Matlab.

Magenta colour of thermal image of the healthy three-phase induction motor is depicted in Fig. 7.

Magenta colour of thermal image of the three-phase induction motor with two broken bars is showed in Fig. 8.

Magenta colour of thermal image of the three-phase induction motor with faulty ring of squirrel-cage is depicted in Fig. 9.

2.2. Method of areas selection of image differences

The MoASoID is developed to compare many training sets of thermal images of electrical motors. A block diagram of the proposed method of feature extraction MoASoID is depicted in Fig. 10. The steps of the proposed method MoASoID are following:

- (1) Calculate differences between images (matrix, 320×240) of magenta colour of the analysed three-phase induction motor $|\mathbf{hi} - \mathbf{i2bb}|$, $|\mathbf{hi} - \mathbf{ifr}|$, $|\mathbf{i2bb} - \mathbf{ifr}|$, where $\mathbf{hi}$ - magenta colour image of the healthy three-phase induction motor, $\mathbf{i2bb}$ - magenta colour image of the three-phase induction motor with 2 broken bars, $\mathbf{ifr}$ - magenta colour image of the three-phase induction motor with faulty ring of squirrel-cage.

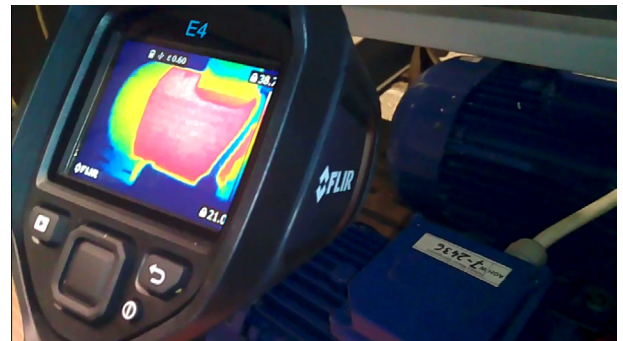


Fig. 3. Thermal imaging camera and the three-phase induction motors.

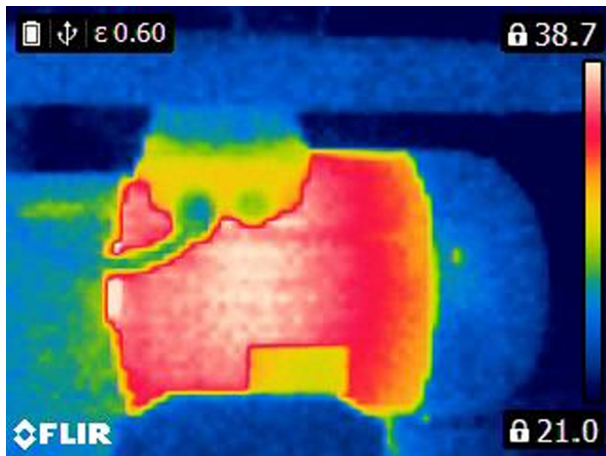


Fig. 4. Thermal image of the healthy three-phase induction motor.

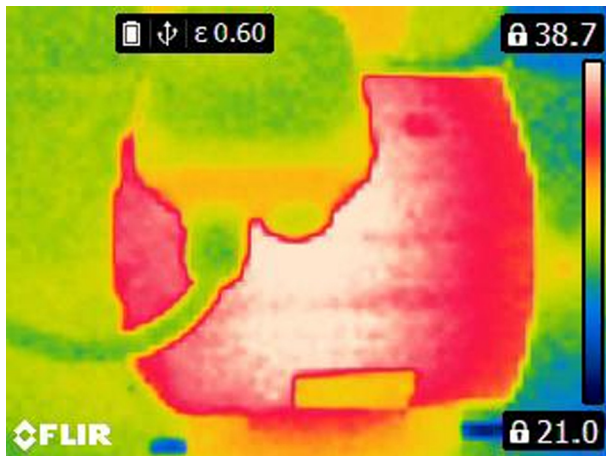


Fig. 5. Thermal image of the three-phase induction motor with two broken bars.

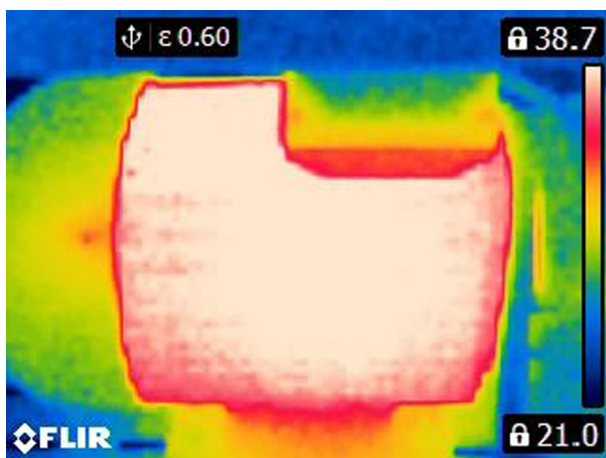


Fig. 6. Thermal image of the three-phase induction motor with faulty ring of squirrel-cage.

- (2) Select the threshold of binarization T ($1 > \text{Threshold} > 0$) for obtained differences. It is essential, because it removes small changes of magenta colour. This parameter is set experimentally. It is analysed in further part of this paper.

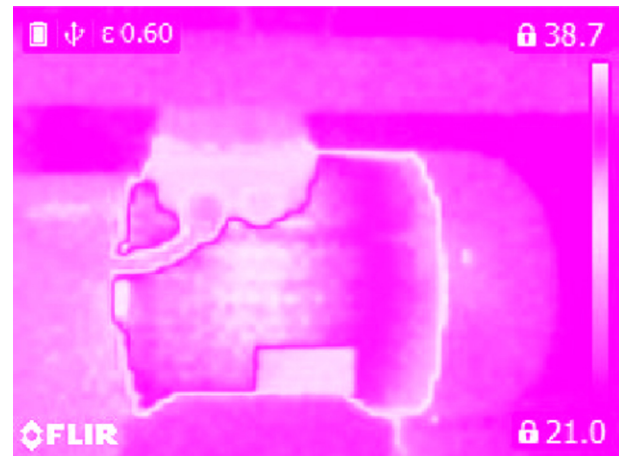


Fig. 7. Magenta colour of thermal image of the healthy three-phase induction motor. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

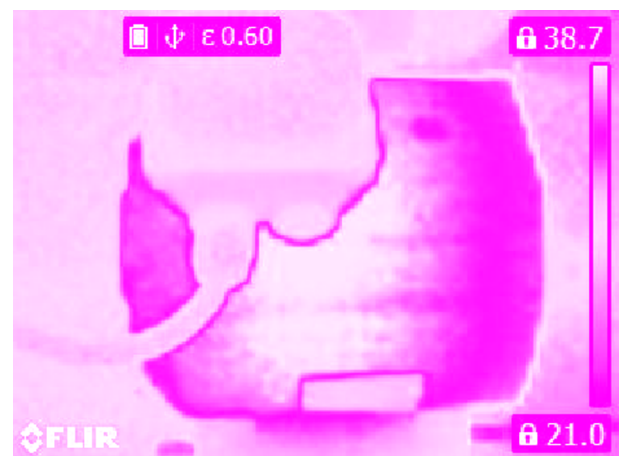


Fig. 8. Magenta colour of thermal image of the three-phase induction motor with two broken bars. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

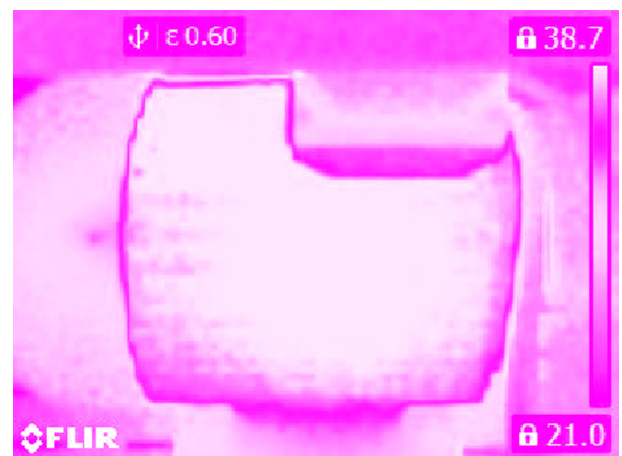


Fig. 9. Magenta colour of thermal image of the three-phase induction motor with faulty ring of squirrel-cage. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

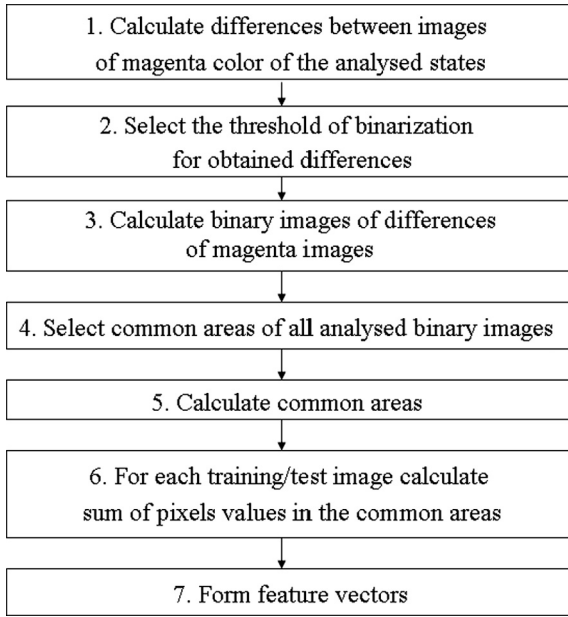


Fig. 10. Block diagram of the MoASoID.

- (3) Calculate binary images of differences of magenta images $|h_i - i2bb|$, $|h_i - ifr|$, $|i2bb - ifr|$.
- (4) Select common areas of all analysed binary images. Each training set has 3 binary images of differences of magenta colour. The authors use 4 training sets (12 binary images) to select common areas.
- (5) Calculate common areas.
- (6) For each training/test image calculate sum of pixels values in the common areas. Sum of pixels values are different for each image. Different values are good for proper recognition.
- (7) Form feature vectors (feature – sum of pixels values).

The differences between images of magenta colour of the three-phase induction motor $|h_i - i2bb|$, $|h_i - ifr|$, $|i2bb - ifr|$ are showed in Figs. 11–13 (rotor speed of 1400 rpm).

Next the common areas of all analysed thermal images are calculated. In the presented analysis 4 training sets are used. Each training set has 3 thermal images. The common areas of 12 anal-



Fig. 11. The difference between magenta colour image of the healthy three-phase induction motor and magenta colour image of the three-phase induction motor with two broken bars ($|h_i - i2bb|$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

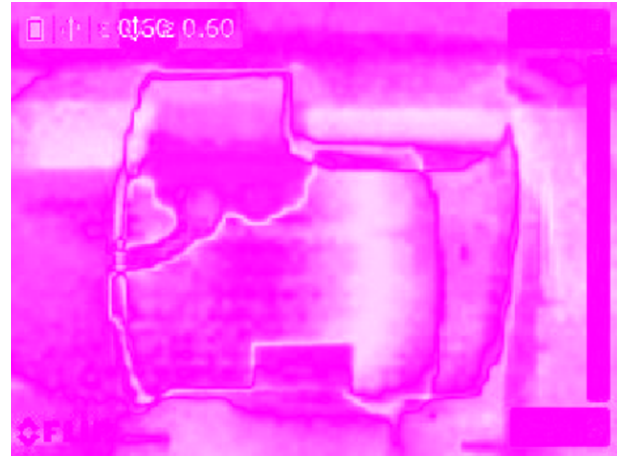


Fig. 12. The difference between magenta colour image of the healthy three-phase induction motor and magenta colour image of the three-phase induction motor with faulty ring of squirrel-cage ($|h_i - ifr|$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

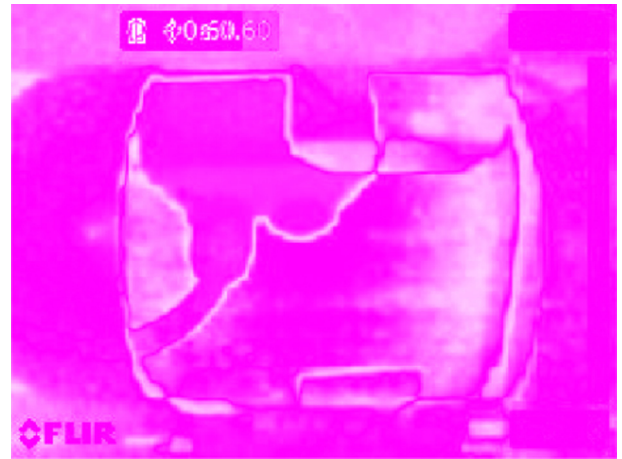


Fig. 13. The difference between magenta colour image of the three-phase induction motor with two broken bars and magenta colour image of the three-phase induction motor with faulty ring of squirrel-cage ($|i2bb - ifr|$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ysed thermal images of the three-phase induction motor are presented in Fig. 14.

The threshold of binarization is based on the analysis. The authors analyse total efficiency of recognition of thermal images TE_R (please see formula (3)). 120 thermal images (40 of each class) are used for the analysis. The MoASoID and the Nearest Neighbour classifier are used to evaluate the total efficiency of recognition of thermal images. It is showed in Fig. 15.

The common areas for three analysed states of the three-phase induction motor are showed in Figs. 16–18.

The obtained sums of pixels values of white colour in the common areas form feature vectors. Feature vectors can be classified by many classifiers in many different ways. The authors use following methods of classification: NN (the Nearest Neighbour classifier) [29], K-means [30], BNN (the back-propagation neural network). Of course other methods of classification such as: decision trees, Naive Bayes classifier [29], SVM (Support Vector Machine) [29,31], GMM (Gaussian Mixture Models) [32,33], fuzzy logic [34], genetic algorithms, self-organizing map [35], neural networks [36–39], LDA (Linear Discriminant Analysis) [40], rough set methods [41] can be also used to solve considered problem.

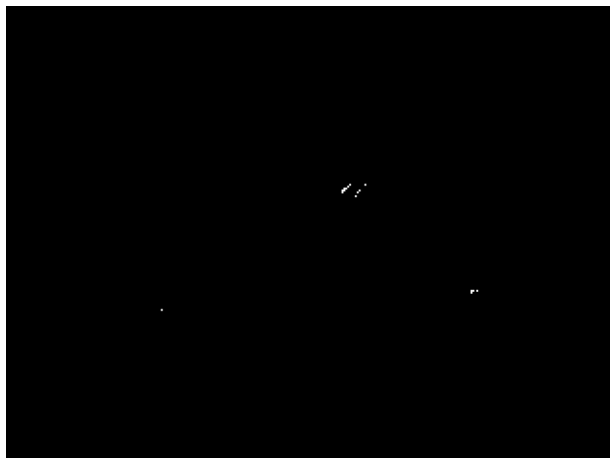


Fig. 14. The common areas of 12 analysed thermal images of the three-phase induction motor. The threshold of binarization is $T = 0.3$.

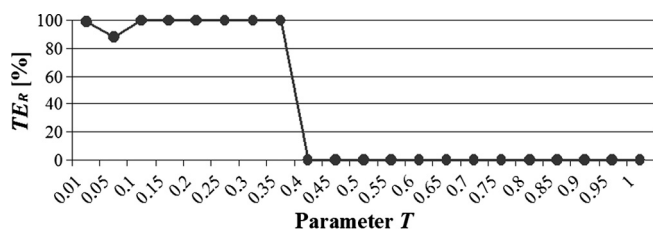


Fig. 15. The total efficiency of recognition of thermal images of three-phase induction motor depending on the threshold of binarization T using the MoASoID and the Nearest Neighbour classifier.

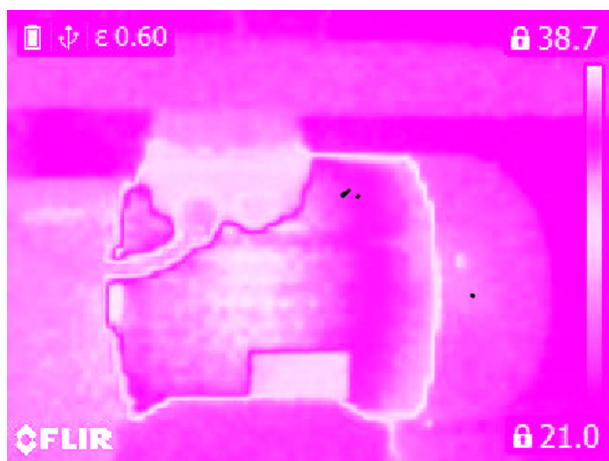


Fig. 16. The common areas of magenta image of the healthy three-phase induction motor. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

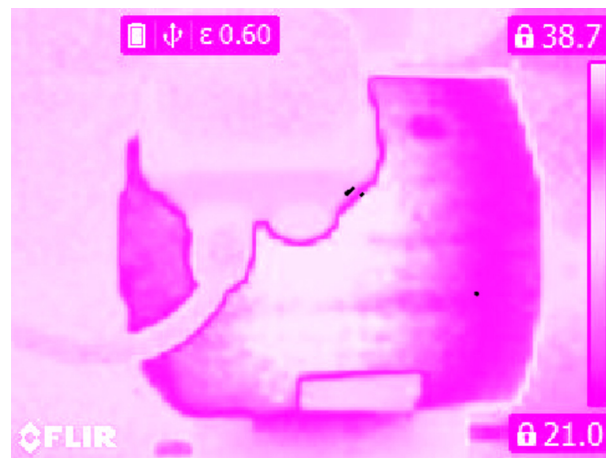


Fig. 17. The common areas of magenta image of the three-phase induction motor with two broken bars. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

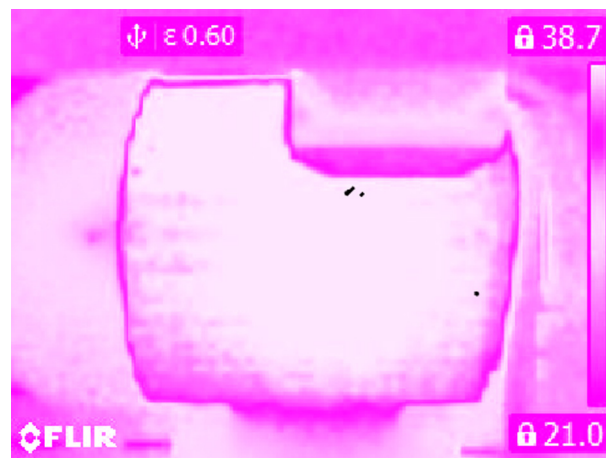


Fig. 18. The common areas of magenta image of the three-phase induction motor with faulty ring of squirrel-cage. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

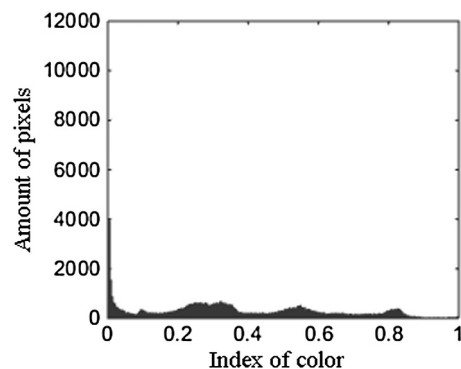


Fig. 19. Image histogram of the healthy three-phase induction motor.

2.3. Feature extraction based on image histogram

The second method of feature extraction is the image histogram. The image histogram is a graphical representation of the tonal distribution of the image. The image histogram was discussed in the literature [42]. Histogram is selected, because of its high recognition efficiency. Image histograms of 3 states of the three-phase induction motor are showed in Figs. 19–21. The pre-

sented histograms are different for 3 states of the three-phase induction motor. The obtained feature vectors have 256 values (index of colour, 0 - black, 1 - white). It is very essential for the next step of recognition - classification.

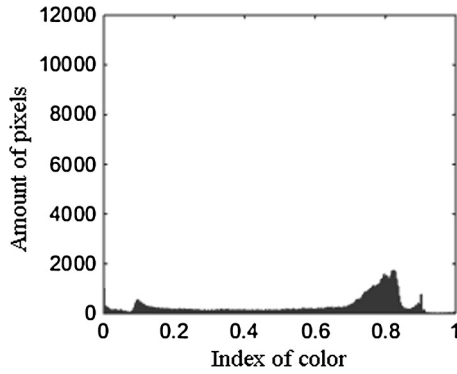


Fig. 20. Image histogram of the three-phase induction motor with two broken bars.

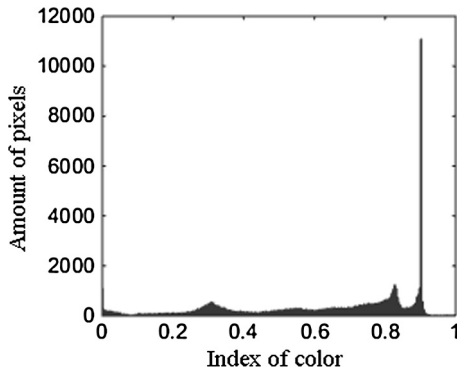


Fig. 21. Image histogram of the three-phase induction motor with faulty ring of squirrel-cage.

2.4. Nearest Neighbour classifier

The Nearest Neighbour classifier was described in many scientific articles [29]. It assigns unclassified test feature vector based on a set of previously classified training feature vectors. This classifier achieves high recognition efficiency, without a priori assumptions about the distributions of training data. It is a reason to use it for fault diagnosis of the three-phase induction motor. The authors use the Manhattan distance for the analysis. This distance function is defined as:

$$d(\mathbf{a}, \mathbf{b}) = \sum_{i=1}^n |a_i - b_i|, \quad (1)$$

where feature vectors are $\mathbf{a} = [a_1, a_2, \dots, a_n]$ and $\mathbf{b} = [b_1, b_2, \dots, b_n]$, $n = 1$ for the MoASoID, $n = 256$ for image histogram.

There are many types of distance function such as: Chebyshev distance, cosine distance, Minkowski distance, Euclidean distance etc. All of them have similar results for obtained feature vectors. In this case the authors decided to analyse the Manhattan distance. More information about the Nearest Neighbour classifier can be found in the literature [29].

2.5. K-means clustering

The one of clustering methods is the K-means. This method is a method of clustering feature vectors into k clusters, where k is the number of clusters. The clustering is done by minimizing the sum of Euclidean/Manhattan distances between training feature vectors and the corresponding centroid. A centroid is the center of mass of a geometric object (geometric object - one cluster). The authors

choose $k = 3$ for analysis, because there are 3 states of the three-phase induction motor (3 acoustic signals).

The K-means clustering has following steps:

1. Set the initial number of centroids and use the k-means++ for centroid initialization.
2. Calculate distances between training feature vectors and centroids.
3. Assign training feature vectors to the corresponding clusters with the nearest centroid.
4. Calculate the arithmetic mean of the training feature vectors in the corresponding clusters to obtain new centroids.
5. Repeat steps 2–4 till new centroids (the arithmetic mean of the training feature vectors) do not change their values.

The K-Means creates centroids in the pattern creation process. Next Manhattan distances between the test feature vector and all centroids are calculated. The authors analyse the Manhattan distance (1), because Euclidean, Minkowski distances give the same results. Next the nearest distance is calculated and the test feature vector is assigned to the corresponding class. More information about the K-Means can be found in the literature [30].

2.6. Back-propagation neural network

Neural networks are related to machine learning and artificial intelligence. Neural networks are the best suited to solving the tasks that are the most difficult to solve by other classification methods. Recognising an object is an image processing task. With neural network technology, we can use processing methods to recognise the object. Neural networks consist of neurons. Neurons receive signals from the neurons in the previous layer. After that obtained signals are multiplied by a weight value. The obtained weighted inputs are summed. Next they are passed through a function. This function scales the output to a fixed range of values. Next the obtained value is sent to all of the neurons in the next layer. The identification process uses test feature vectors. Next these feature vectors are applied to the inputs of the first layer. Next the signals propagate through the network. After that the output values are obtained in the output layer.

A back-propagation neural network learns by training feature vectors. A training set consists of training feature vectors and the known-correct output for each case. The learning process of the back propagation neural network has some steps. The first of them is the application of training feature vectors to the neural network. The neural network produces some output based on the current state of its weights and training feature vectors. Next obtained output is compared to the known-correct output. The back-propagation algorithm computes a mean-squared error. After that the computed error value is propagated backwards through the network. Small changes are computed to the weights in each layer. The weight changes are computed to decrease the mean-squared error. The learning process is repeated for each of training feature vector. It is repeated until the total error value dropped below threshold. The neural network is learnt “well” for this value of threshold. More information about the neural network can be found in the literature [36–39].

The authors use the following back-propagation neural network (Fig. 22). Output values are following: 0001 - class 1 (healthy three-phase induction motor), 0010 - class 2 (three-phase induction motor with 2 broken bars), 0100 - class 3 (three-phase induction motor with faulty ring of squirrel-cage) and 1000 - class 4 (unknown).

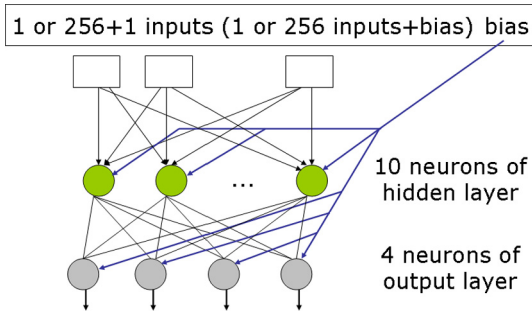


Fig. 22. Structure of the analysed back-propagation neural network.

3. Analysis of the results

In the proposed analysis thermal images are measured under laboratory conditions. Three the same induction motors are investigated. Motors operate under open loop control. The parameters of the analysed machines are: $P = 550$ W, $r_s = 1400$ rpm, $I_{SC} = 2.52/1.47$ A (Δ/Y), $U_{SV} = 220/380$ V (Δ/Y), where P - motor power, r_s - rotor speed, I_{SC} - nominal stator current, U_{SV} - nominal stator voltage. The authors analyse thermal images of 3 states of the three-phase induction motor: healthy three-phase induction motor, three-phase induction motor with 2 broken bars (Fig. 23), three-phase induction motor with faulty ring of squirrel-cage (Fig. 24).

The pattern creation process is carried out for 12 training images. The identification process is carried out for 120 test images. Training and test images are processed by proposed the diagnostic technique (Fig. 2). The feature vectors are computed by the MoASoID. After that they are classified by NN, K-means, BNN. The efficiency of recognition of thermal images is expressed as (2):

$$E_R = \frac{N_{CCTFV}}{N_{ATFV}} \cdot 100\% \quad (2)$$

where E_R - the efficiency of recognition of thermal image, N_{CCTFV} - the number of correctly classified test feature vectors, N_{ATFV} - the number of all test feature vectors.

To compare the results of recognition, the authors introduce the total efficiency of recognition of thermal images. It is defined as (3):

$$TE_R = \frac{E_{RH} + E_{R2B} + E_{RFR}}{3} \quad (3)$$

where TE_R - the total efficiency of recognition of thermal image, E_{RH} - the efficiency of recognition of thermal image of the healthy three-phase induction motor, E_{R2B} - the efficiency of recognition of thermal image of the three-phase induction motor with 2 broken bars, E_{RFR} - the efficiency of recognition of thermal image of the three-phase induction motor with faulty ring of squirrel-cage.

The obtained results are showed in Tables 1–4. In Table 1 the authors present the results of recognition using the MoASoID and the Nearest Neighbour classifier.

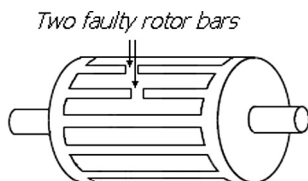


Fig. 23. Rotor of the three-phase induction motor with two broken bars.

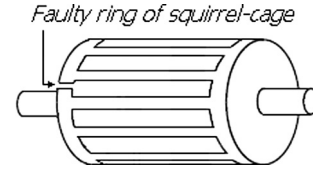


Fig. 24. Rotor of the three-phase induction motor with faulty ring of squirrel-cage.

Table 1

The results of recognition of thermal images of the three-phase induction motor using the MoASoID and the Nearest Neighbour classifier.

Type of thermal image	E_R [%]
Healthy three-phase induction motor	100
Three-phase induction motor with 2 broken bars	100
Three-phase induction motor with faulty ring of squirrel-cage	100
TE_R	100

In Table 2 the authors present the results of recognition using the MoASoID and the K-means.

In Table 3 the authors present the results of recognition using the MoASoID and the back-propagation neural network.

In Table 4 the authors present the results of recognition using image histogram and the Nearest Neighbour classifier.

The obtained results are very good. The value of E_R and the value of TE_R are equal to 100%. Feature extraction methods (the MoASoID and the image histogram) are good for recognition of thermal images of the analysed motor. The applied classifiers are also very efficient. The developed fault diagnostic techniques are able to classify thermal images of the three-phase induction motor correctly.

Table 2

The results of recognition of thermal images of the three-phase induction motor using the MoASoID and the K-means.

Type of thermal image	E_R [%]
Healthy three-phase induction motor	100
Three-phase induction motor with 2 broken bars	100
Three-phase induction motor with faulty ring of squirrel-cage	100
TE_R	100

Table 3

The results of recognition of thermal images of the three-phase induction motor using the MoASoID and the back-propagation neural network.

Type of thermal image	E_R [%]
Healthy three-phase induction motor	100
Three-phase induction motor with 2 broken bars	100
Three-phase induction motor with faulty ring of squirrel-cage	100
TE_R	100

Table 4

Results of recognition of thermal images of the three-phase induction motor using image histogram and the Nearest Neighbour classifier.

Type of thermal image	E_R [%]
Healthy three-phase induction motor	100
Three-phase induction motor with 2 broken bars	100
Three-phase induction motor with faulty ring of squirrel-cage	100
TE_R	100

4. Conclusions

In this paper the authors develop fault diagnostic techniques of the three-phase induction motor. The described techniques are based on the analysis of thermal images of three-phase induction motor. The authors analyse thermal images of 3 states of the three-phase induction motor: healthy three-phase induction motor, three-phase induction motor with 2 broken bars, three-phase induction motor with faulty ring of squirrel-cage. In this paper the authors develop the original method of the feature extraction MoASoID. The feature vectors are obtained with the use of mentioned MoASoID and image histogram. The classification methods are used: NN, K-means, BNN. The obtained results are very good. The value of E_R and the value of TE_R are equal to 100%. In authors' opinion, the best choice is the Nearest Neighbour classifier. This classifier has less errors than the others classifiers. The developed fault diagnostic techniques are able to classify thermal images of the three-phase induction motor properly.

The first advantage of proposed technique is that, it is non-invasive. It is conducted that healthy three-phase induction motor, three-phase induction motor with 2 broken bars, three-phase induction motor with faulty ring of squirrel-cage can be detected using proposed technique. The described technique is implemented in Matlab language and it works fast (2 s for identification process of test sample on PC computer, processor i7-4702MQ). Next advantage is related to the method of feature extraction MoASoID. This method compares many training sets together and it selects the areas with the biggest changes for the recognition process. The described fault diagnostic techniques are useful for protection of three-phase induction motor and other types of rotating electrical motors such as: DC motors, generators, synchronous motors.

However there are some disadvantages. The first disadvantage of this method is the cost of thermal imaging camera. It costs about 1700\$, so it is more expensive than analysis of acoustic signals. The second disadvantage is that, average time to heat up stator coils is about 1–3 min, for example vibrations and acoustic signals can be measured earlier. Moreover long durations of shorted coils can cause permanent damage to the analysed machine. Thermal imaging camera can be set in many ways, so the training and test sets of thermal images may be improper for recognition process.

In the future, other signals such as: electric current, electric voltage, vibration, acoustic signal, magnetic should be also analysed. These tasks are required to develop more reliable fault diagnostic techniques of electrical motors.

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References

- [1] H. Henao, G.A. Capolino, M. Fernandez-Cabanas, F. Filippetti, C. Bruzzese, E. Strangas, R. Pusca, J. Estima, M. Riera-Guasp, S. Hedayati-Kia, Trends in fault diagnosis for electrical machines a review of diagnostic techniques, *IEEE Ind. Electron. Mag.* 8 (2) (2014) 31–42.
- [2] A.G. Garcia-Ramirez, L.A. Morales-Hernandez, R.A. Osornio-Rios, J.P. Benitez-Rangel, A. Garcia-Perez, R.D. Romero-Troncoso, Fault detection in induction motors and the impact on the kinematic chain through thermographic analysis, *Electr. Power Syst. Res.* 114 (2014) 1–9, <http://dx.doi.org/10.1016/j.epsr.2014.03.031>.
- [3] Khalaf Salloum Gaied, Hew Wooi Ping, Mustafa Khalid, Atheer Laury Salih, Fault diagnosis of induction motor using MCSA and FFT, *Electr. Electron. Eng.* 1 (2) (2011) 85–92, <http://dx.doi.org/10.5923/j.eee.20110102.14>.
- [4] T. Ciszewski, L. Gelman, L. Swedrowski, Current-based higher-order spectral covariance as a bearing diagnostic feature for induction motors, *Insight* 58 (8) (2016) 431–434, <http://dx.doi.org/10.1784/insi.2016.58.8.431>.
- [5] M. Sulowicz, K. Weinreb, R. Mielnik, T. Zywczak, M. Jaraczewski, The method of current measurement in the rotor cage bars of prototype induction motor with the use of Rogowski coils, in: *International Conference on Information and Digital Technologies (IDT)*, 2015, pp. 357–365.
- [6] M. Michalak, B. Sikora, J. Sobczyk, Diagnostic model for longwall conveyor engines, in: *Man-Machine Interactions 4, ICMMI 2015, Book Series: Advances in Intelligent Systems and Computing*, 391, 2016, pp. 437–448, http://dx.doi.org/10.1007/978-3-319-23437-3_37.
- [7] A.G. Garcia-Ramirez, L.A. Morales-Hernandez, R.A. Osornio-Rios, A. Garcia-Perez, R.J. Romero-Troncoso, Thermographic technique as a complement for MCSA in induction motor fault detection, in: *2014 International Conference on Electrical Machines (ICEM)*, 2014, pp. 1940–1945.
- [8] P.A. Delgado-Arredondo, A. Garcia-Perez, D. Morinigo-Sotelo, R.A. Osornio-Rios, J.G. Avina-Cervantes, H. Rostro-Gonzalez, R.D. Romero-Troncoso, Comparative study of time-frequency decomposition techniques for fault detection in induction motors using vibration analysis during startup transient, *Shock Vib.* (2015), <http://dx.doi.org/10.1155/2015/708034> Article Number: 708034.
- [9] J.J. Saucedo-Dorantes, M. Delgado-Prieto, J.A. Ortega-Redondo, R.A. Osornio-Rios, R.D. Romero-Troncoso, Multiple-fault detection methodology based on vibration and current analysis applied to bearings in induction motors and gearboxes on the kinematic chain, *Shock Vib.* (2016), <http://dx.doi.org/10.1155/2016/5467643> Article Number: 5467643.
- [10] Z.X. Li, Y. Jiang, C. Hu, Z. Peng, Recent progress on decoupling diagnosis of hybrid failures in gear transmission systems using vibration sensor signal: a review, *Measurement* 90 (2016) 4–19, <http://dx.doi.org/10.1016/j.measurement.2016.04.036>.
- [11] Z.X. Li, X.P. Yan, X.P. Wang, Z.X. Peng, Detection of gear cracks in a complex gearbox of wind turbines using supervised bounded component analysis of vibration signals collected from multi-channel sensors, *J. Sound Vib.* 371 (2016) 406–433, <http://dx.doi.org/10.1016/j.jsv.2016.02.021>.
- [12] A. Glowacz, Z. Glowacz, Diagnosis of stator faults of the single-phase induction motor using acoustic signals, *Appl. Acoust.* 117 (Part A) (2017) 20–27, <http://dx.doi.org/10.1016/j.apacoust.2016.10.012>.
- [13] A. Glowacz, Fault diagnostics of acoustic signals of loaded synchronous motor using SMOFS-25-EXPANDED and selected classifiers, *Tehnicky Vjesnik – Tech. Gaz.* 23 (5) (2016) 1365–1372.
- [14] A. Deptula, P. Osinski, U. Radziwanowska, Decision support system for identifying technical condition of combustion engine, *Arch. Acoust.* 41 (3) (2016) 449–460.
- [15] A. Deptula, D. Kunderman, P. Osinski, U. Radziwanowska, R. Wlostowski, Acoustic diagnostics applications in the study of technical condition of combustion engine, *Arch. Acoust.* 41 (2) (2016) 345–350, <http://dx.doi.org/10.1515/aoa-2016-0036>.
- [16] P.A. Delgado-Arredondo, D. Morinigo-Sotelo, R.A. Osornio-Rios, J.G. Avina-Cervantes, H. Rostro-Gonzalez, R.D. Romero-Troncoso, Methodology for fault detection in induction motors via sound and vibration signals, *Mech. Syst. Signal Process.* 83 (2017) 568–589, <http://dx.doi.org/10.1016/j.ymssp.2016.06.032>.
- [17] B. Van Hecke, J. Yoon, D. He, Low speed bearing fault diagnosis using acoustic emission sensors, *Appl. Acoust.* 105 (2016) 35–44, <http://dx.doi.org/10.1016/j.apacoust.2015.10.028>.
- [18] M. Sebok, M. Gutten, M. Kucera, Diagnostics of electric equipments by means of thermovision, *Przegląd Elektrotechniczny* 87 (10) (2011) 313–317.
- [19] J.A.R. Nunez, L.M. Velazquez, L.A.M. Hernandez, R.J.R. Troncoso, R.A. Osornio-Rios, Low-cost thermographic analysis for bearing fault detection on induction motors, *J. Sci. Ind. Res.* 75 (7) (2016) 412–415.
- [20] G. Singh, T.C.A. Kumar, V.N.A. Naikan, Induction motor inter turn fault detection using infrared thermographic analysis, *Infrared Phys. Technol.* 77 (2016) 277–282, <http://dx.doi.org/10.1016/j.infrared.2016.06.010>.
- [21] D.K. Chaturvedi, M.S. Iqbal, M.P. Singh, Intelligent health monitoring system for three phase induction motor using infrared thermal image, in: *International Conference on Energy Economics and Environment (ICEEE)* 2015, Greater Noida, India, 2015.
- [22] T.B. Bai, L.B. Zhang, L.X. Duan, J.J. Wang, NSCT-based infrared image enhancement method for rotating machinery fault diagnosis, *IEEE Trans. Instrum. Meas.* 65 (10) (2016) 2293–2301, <http://dx.doi.org/10.1109/TIM.2016.2579440>.
- [23] Y. Xie, Y.Y. Wang, 3D temperature field analysis of the induction motors with broken bar fault, *Appl. Therm. Eng.* 66 (1–2) (2014) 25–34, <http://dx.doi.org/10.1016/j.applthermaleng.2014.02.008>.
- [24] B. Gao, W.L. Woo, Y.Z. He, G.Y. Tian, Unsupervised sparse pattern diagnostic of defects with inductive thermography imaging system, *IEEE Trans. Ind. Inf.* 12 (1) (2016) 371–383, <http://dx.doi.org/10.1109/TII.2015.2492925>.
- [25] H. Liu, Z.X. Wang, J. Zhong, Z.W. Xie, Early detection of spontaneous combustion disaster of sulphide ore stockpiles, *Tehnicky Vjesnik – Tech. Gaz.* 22 (6) (2015) 1579–1587.
- [26] J.P. Bazzo, F. Mezzadri, E.V. da Silva, D.R. Pipa, J.C.C. da Silva, Thermal imaging of hydroelectric generator stator using a DTS system, *IEEE Sens. J.* 15 (11) (2015) 6689–6696, <http://dx.doi.org/10.1109/JSEN.2015.2465862>.
- [27] H. Glavas, L. Jozsa, T. Baric, Infrared thermography in energy audit of electrical installations, *Tehnicky Vjesnik – Tech. Gaz.* 23 (5) (2016) 1533–1539, <http://dx.doi.org/10.17559/TV-20150702185559>.
- [28] M. Marzec, R. Koprowski, Z. Wrobel, Methods of face localization in thermograms, *Biocybern. Biomed. Eng.* 35 (2) (2015) 138–146, <http://dx.doi.org/10.1016/j.bbe.2014.09.001>.

- [29] R.H.C. Palacios, I.N. da Silva, A. Goedel, W.F. Godoy, A comprehensive evaluation of intelligent classifiers for fault identification in three-phase induction motors, *Electr. Power Syst. Res.* 127 (2015) 249–258, <http://dx.doi.org/10.1016/j.epsr.2015.06.008>.
- [30] D. Camarena-Martinez, M. Valtierra-Rodriguez, J.P. Amezcua-Sanchez, D. Granados-Lieberman, R.J. Romero-Troncoso, A. Garcia-Perez, Shannon entropy and k-means method for automatic diagnosis of broken rotor bars in induction motors using vibration signals, *Shock Vib.* (2016), <http://dx.doi.org/10.1155/2016/4860309> Article Number: 4860309.
- [31] A. Swetapadma, A. Yadav, Directional relaying using support vector machine for double circuit transmission lines including cross-country and inter-circuit faults, *Int. J. Electr. Power Energy Syst.* 81 (2016) 254–264, <http://dx.doi.org/10.1016/j.ijepes.2016.02.034>.
- [32] E.P. Frigieri, P.H.S. Campos, A.P. Paiva, P.P. Balestrassi, J.R. Ferreira, C.A. Ynoguti, A mel-frequency cepstral coefficient-based approach for surface roughness diagnosis in hard turning using acoustic signals and gaussian mixture models, *Appl. Acoust.* 113 (2016) 230–237, <http://dx.doi.org/10.1016/j.apacoust.2016.06.027>.
- [33] R.M. Bowen, F. Sahin, A. Radomski, Systemic health evaluation of RF generators using Gaussian mixture models, *Comput. Electr. Eng.* 53 (2016) 13–28, <http://dx.doi.org/10.1016/j.compeleceng.2016.04.020>.
- [34] D. Valis, K. Pietrucha-Urbanik, Utilization of diffusion processes and fuzzy logic for vulnerability assessment, *Eksplotacja i Niezawodnosc – Maint. Reliab.* 16 (1) (2014) 48–55.
- [35] D. Jamroz, T. Niedoba, Application of multidimensional data visualization by means of self-organizing Kohonen maps to evaluate classification possibilities of various coal types, *Arch. Min. Sci.* 60 (1) (2015) 39–50, <http://dx.doi.org/10.1515/amsc-2015-0003>.
- [36] R.H.C. Palacios, A. Goedel, W.F. Godoy, J.A. Fabri, Fault identification in the stator winding of induction motors using PCA with artificial neural networks, *J. Control Autom. Electr. Syst.* 27 (4) (2016) 406–418, <http://dx.doi.org/10.1007/s40313-016-0248-0>.
- [37] S. Jun, O. Kochan, Investigations of thermocouple drift irregularity impact on error of their inhomogeneity correction, *Meas. Sci. Rev.* 14 (1) (2014) 29–34.
- [38] R. Tadeusiewicz, Neural networks in mining sciences—general overview and some representative examples, *Arch. Min. Sci.* 60 (4) (2015) 971–984, <http://dx.doi.org/10.1515/amsc-2015-0064>.
- [39] L. Jedlinski, J. Caban, L. Krzywonos, S. Wierzbicki, F. Brumerick, Application of vibration signal in the diagnosis of IC engine valve clearance, *J. Vibroeng.* 17 (1) (2015) 175–187.
- [40] C.P. Mbo'o, K. Hameyer, Fault diagnosis of bearing damage by means of the linear discriminant analysis of stator current features from the frequency selection, *IEEE Trans. Ind. Appl.* 52 (5) (2016) 3861–3868, <http://dx.doi.org/10.1109/TIA.2016.2581139>.
- [41] Z. Gorny, S. Kluska-Nawarecka, D. Wilk-Kolodziejczyk, K. Regulski, Methodology for the construction of a rule-based knowledge base enabling the selection of appropriate bronze heat treatment parameters using rough sets, *Arch. Metall. Mater.* 60 (1) (2015) 309–312, <http://dx.doi.org/10.1515/amm-2015-0050>.
- [42] E. Kucukkulahli, P. Erdogmus, K. Polat, Histogram-based automatic segmentation of images, *Neural Comput. Appl.* 27 (5) (2016) 1445–1450, <http://dx.doi.org/10.1007/s00521-016-2287-7>.
- [43] A. Glowacz, Z. Glowacz, Diagnostics of stator faults of the single-phase induction motor using thermal images, MoASoS and selected classifiers, *Measurement* 93 (2016) 86–93, <http://dx.doi.org/10.1016/j.measurement.2016.07.008>.